

Satyabrata Das

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SUMMARY

MS AI student at the University of Florida specializing in computer vision, deep learning, and real-time AI systems. Experienced in building scalable ML pipelines, GPU-accelerated inference systems, and production-grade AI applications using PyTorch, OpenCV, FastAPI, and Docker.

EDUCATION

University of Florida

M.S., Artificial Intelligence Systems

Aug 2025 - May 2027

Gainesville, FL

- **Coursework:** AI Safety & Robustness, AI Systems, Machine Learning for AI, Image Processing & Computer Vision, Applied Deep Learning

EXPERIENCE

Evy

May 2026 - Present

Computer Vision Engineer – Eye Tracking & ML

Florida, USA

- Built real-time eye tracking and blink detection systems for AI interview monitoring using Python, OpenCV, and MediaPipe; improved gaze estimation accuracy and reduced false positives across varied lighting, webcams, and user environments.
- Prototyped PyTorch and TensorFlow models that classified user behavior from gaze and blink data and implemented anomaly detection, enabling the system to flag suspicious interview patterns with higher reliability
- Optimized PyTorch inference pipelines to reduce latency, integrated browser-based computer-vision using WebGL, and worked with founders to launch the solution, resulting in a deployment that supports multiple concurrent users

ARC Document Solutions

Jan 2022 - Sep 2024

Software Engineer

India

- Built scalable real-time applications with Swift, WebSockets, and REST APIs for production AI platforms, enabling faster data processing and improving user response times.
- Developed automation and backend-integrated workflows supporting scalable application performance and reliability.

SCI-BI Software Pvt Ltd

Jul 2021 - Dec 2021

Software Trainee

India

- Developed ETL and automated data-processing pipelines handling millions of records daily with 99%+ accuracy.
- Optimized SQL workflows and set up Bash scripts with Prometheus and Grafana monitoring, improving data availability and reducing processing latency for large-scale datasets

PROJECTS

Poselnsight – Real-Time Movement Analysis [GitHub](#)

- Built real-time pose estimation pipelines (OpenCV, MediaPipe) and trained LSTM-based movement-quality classification models; optimized inference latency via asynchronous frame processing and containerized deployment.

JuggleIQ – CV-Based Soccer Analytics [GitHub](#)

- Architected GPU-accelerated CV pipelines (1,000+ frame-level inferences/session) integrating YOLO, pose estimation, and Kalman filtering; FastAPI + Docker.

SummarIQ – Math-Aware Research Paper Summarizer [GitHub](#)

- Built transformer-based (HuggingFace T5/BART) summarization with LaTeX extraction for 200+ papers; GPU-accelerated FastAPI deployment with sub-1.4s latency.

SKILLS

- **AI & ML:** PyTorch, TensorFlow Lite, HuggingFace Transformers, Scikit-learn, LLMs, NLP, CUDA
- **Computer Vision:** OpenCV, MediaPipe, YOLO, 3D Pose Estimation, Eye Tracking, Gaze Estimation, Video Analytics
- **Generative AI & Retrieval:** RAG, Vector Databases, Embedding Models, Retrieval Systems
- **Programming:** Python, C++, Swift, SQL
- **Systems & Tools:** FastAPI, Docker, REST APIs, CI/CD, Git, Linux, Prometheus/Grafana, Weights & Biases

ACHIEVEMENTS & CERTIFICATIONS

- **Winner:** NASA EMERGE Data Hackathon

CERTIFICATION

- NVIDIA: Fundamentals of Deep Learning; Building Transformer-Based NLP Applications
- DeepLearning.AI: Supervised ML